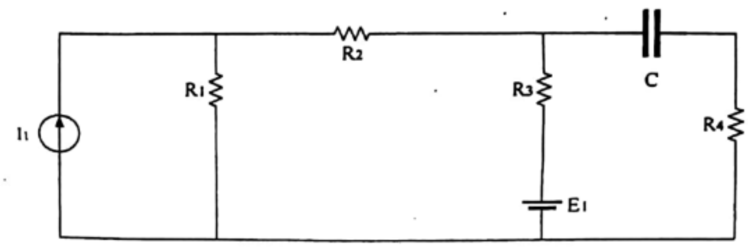
UNIT 1 DC NETWORKS

Assume suitable missing data, if any.

Q.1 Answer the following :

(i) Indicate the series – parallel connections of individual elements in the circuit of Fig.1, as it is. **[2]**



R2 and R3 are in series
R23 and R1 in parallel

Fig. 1

(ii) Is superposition principle applicable on the following functions? Explain. **[2]**

a) $y = 5x^2+2x+7$

b) $y = \text{Sin } x$

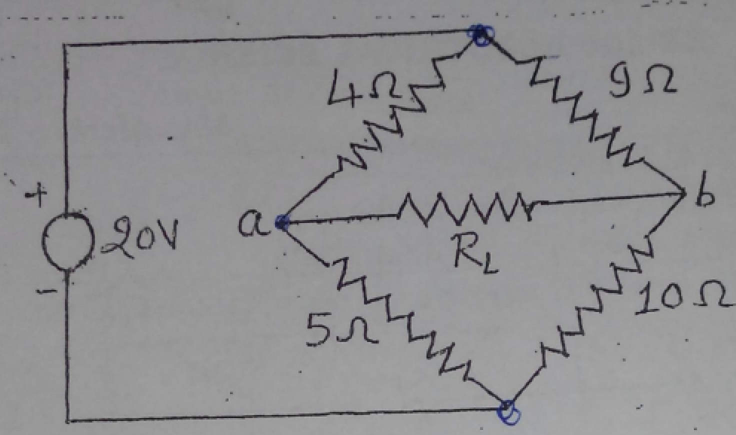
NOT APPLICABLE

check if the function is a linear function or not.
i.e. f(x1 + x2) = f(x1) + f(x2)

(iii) What are the various types of dependent sources. Draw the symbol of each. **[2]**

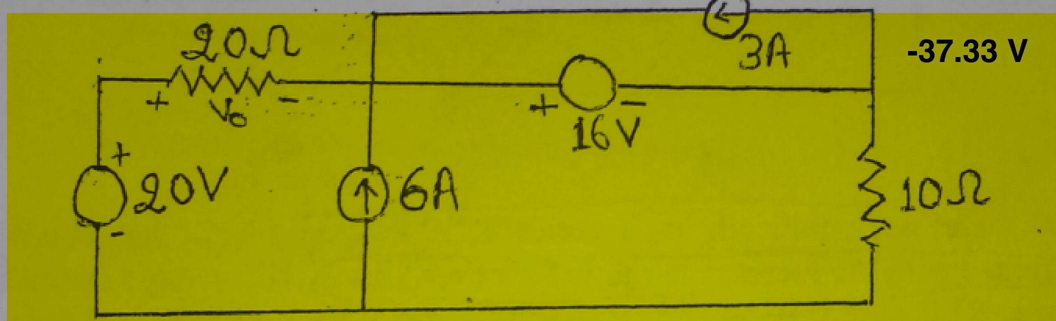
Q.3 Calculate the value of Load Resistance R_L for which maximum power is transferred to the load resistor R_L. Also calculate the maximum power transferred to R_L. (5)

6.95 ohm
0.0121 W

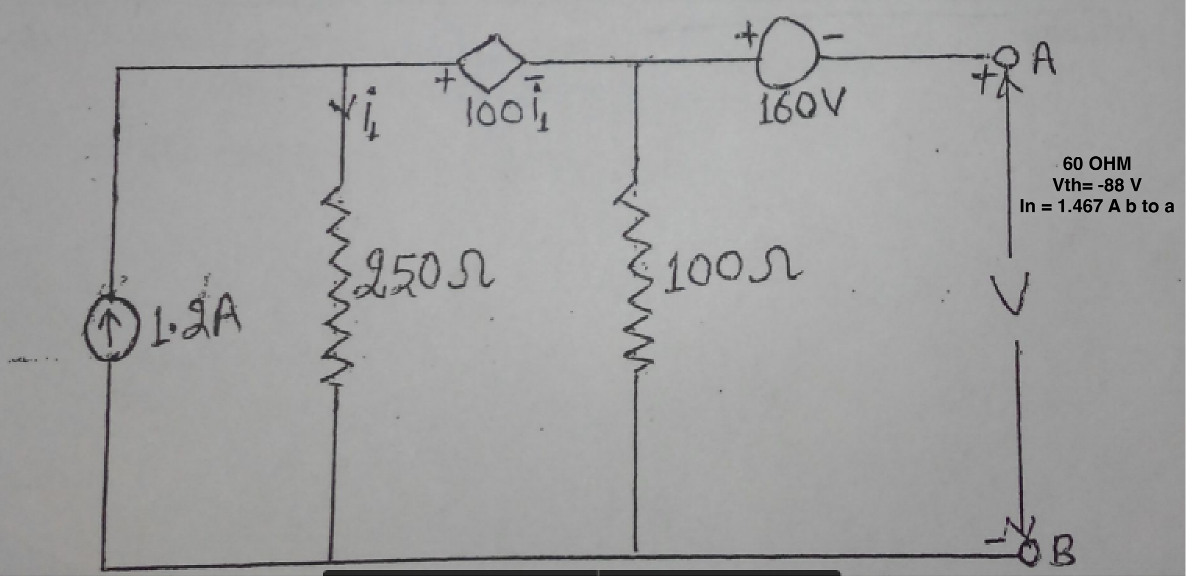


Assume suitable missing data, if any.

Q.1 Find voltage across 20Ω resistor using superposition theorem. (5)



Q.2 Find the Norton's & Thevenin's Models between terminal 'A' and 'B' of the given circuit diagram. (5)



5. Define the following terms:

(a) Quality factor (b) Form factor (c) Bandwidth (d) Resonance [4]

6. State and explain superposition theorem with the help of suitable example. [4]

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4. What is the potential difference between X and Y of the network shown in Fig. 4. [4]

-1.2 V

0.2

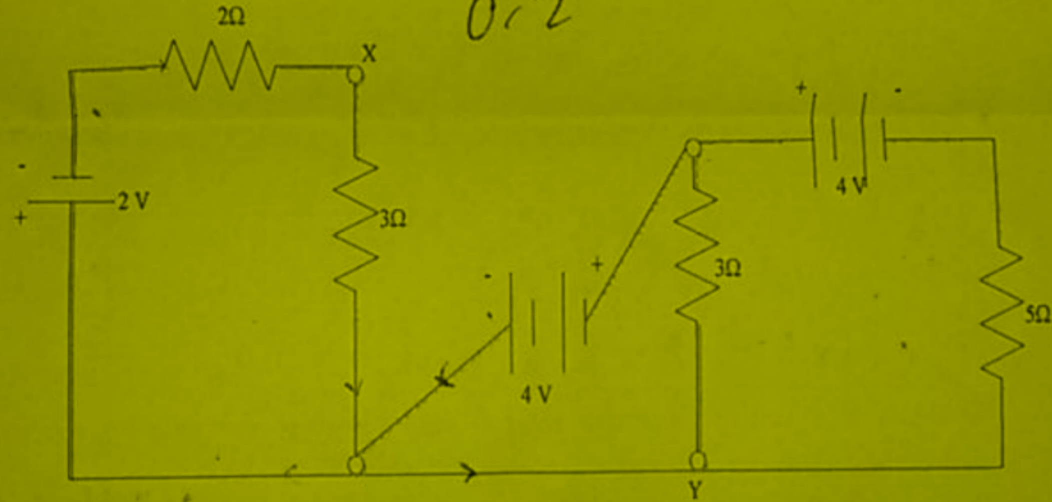


Fig. 4

1. *Find the Thevenin's equivalent circuit of the network shown in Fig. 1.* [4]

$\omega = \frac{1}{\sqrt{LC}}$

$C = \frac{1}{\sqrt{L}\,\omega}$

$= \frac{1}{L\omega^2}$

$= \frac{1}{0.01 \times 4\pi^2 \times 50 \times 50}$

Vth = 0
Rth = 0.6 ohm

6.67, 1.2

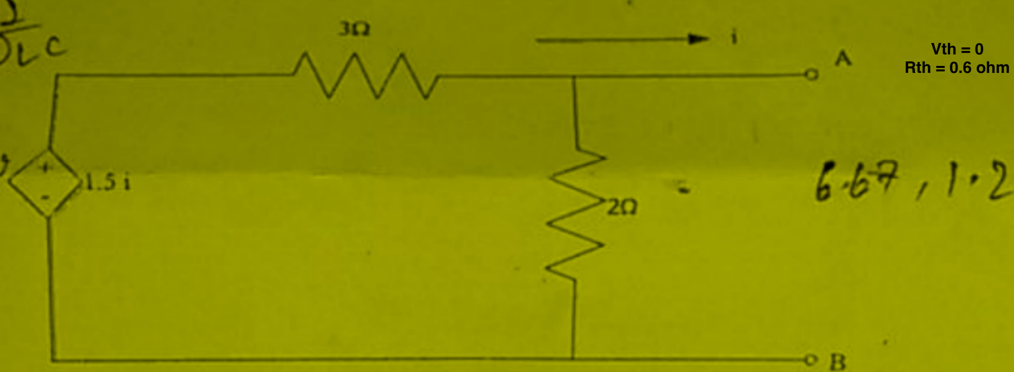
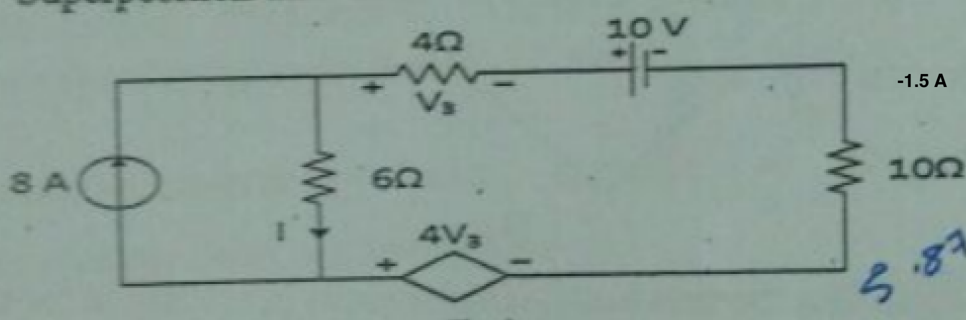


Fig. 1.

Q.1. Find the current I in the circuit shown in Fig.1 using Superposition Theorem.



-1.5 A

5.87

Fig.1

Q.2. Determine the current I in the network of Fig. 2 using Y- Δ and Δ–Y transformations.

10 A

10 A

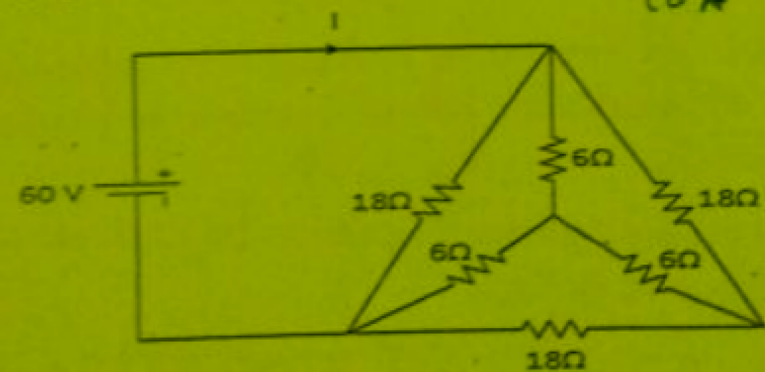


Fig.2

Fig. 4

Q. 5 State whether principle of superposition is applicable to the following functions and justify your answer. **[5]**

i. $y = 5x_1 + 2x_2 + 7x_3$

ii. $y = 7x^2 + 7$

iii. $y = \cos x_1 + \sin x_2$

iv. $y = 2x$

v. $y = 2x + 7$

Applicable only for IV

Q.6 If AB, CD represent two resistances R_1 and R_2 show that EF represents, to the same scale, the value of resistance R_1 and R_2 in parallel. Find graphically, and check by calculation the resistance between X and Y in the network shown in Fig. 5. **[5]**

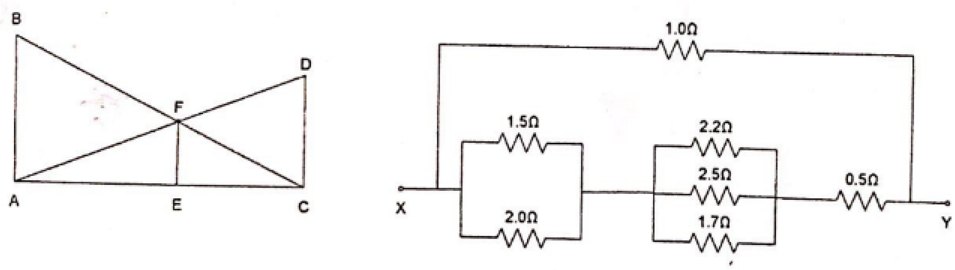
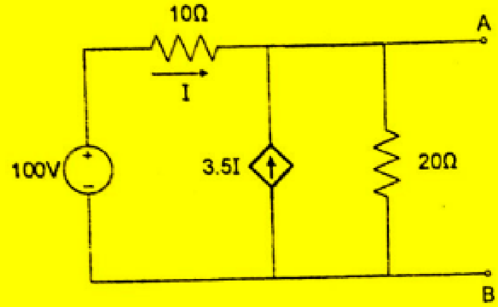


Fig. 5

OR

A star network in which N is the star point is made up as follows: AN = 70 Ω, BN= 100 Ω and CN = 90 Ω. Find an equivalent delta network.
If the above star-delta network is superimposed, what would be the resistance measured between A and C?

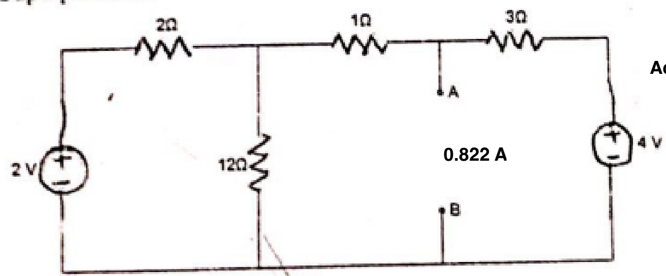
Q.1 Determine Thevenin equivalent of the circuit shown in Fig. 1. **[5]**



2 ohm and 90 V

Fig. 1

Q. 2 Find the current in 2Ω resistor connected between A and B in the circuit shown in Fig. 2 using Principle of Superposition. **[5]**

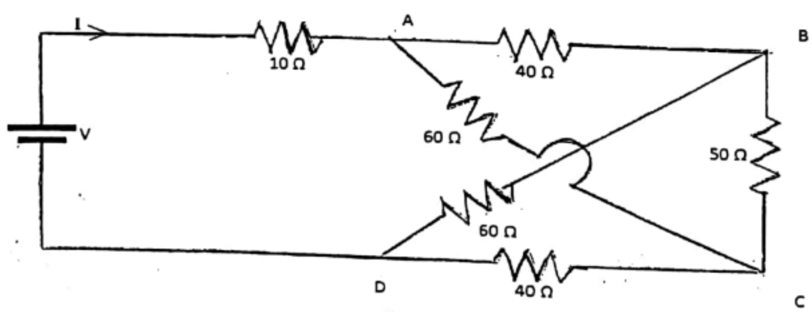


HInt:
Add a 2 ohm resistor across AB and solve .

0.822 A

Fig. 2

Q. 3 For the network shown in Fig.3, use star – delta transformation to find voltage V that makes current I=3A. **[4]**



176.87 V

Fig. 3

Q.4 In the circuit of Fig. 4, determine the value of resistor R_L that will absorb maximum power and what is the value of this power?

4.8 ohm
270 W

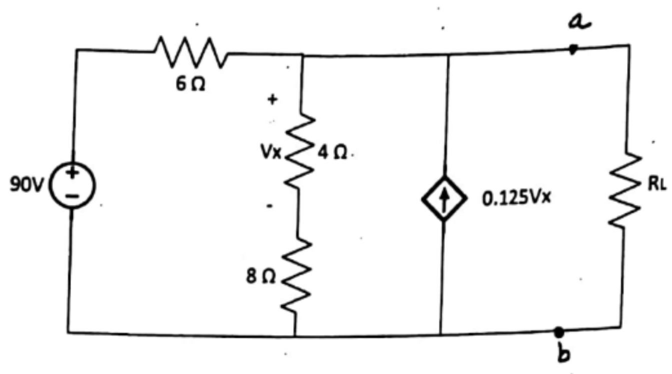


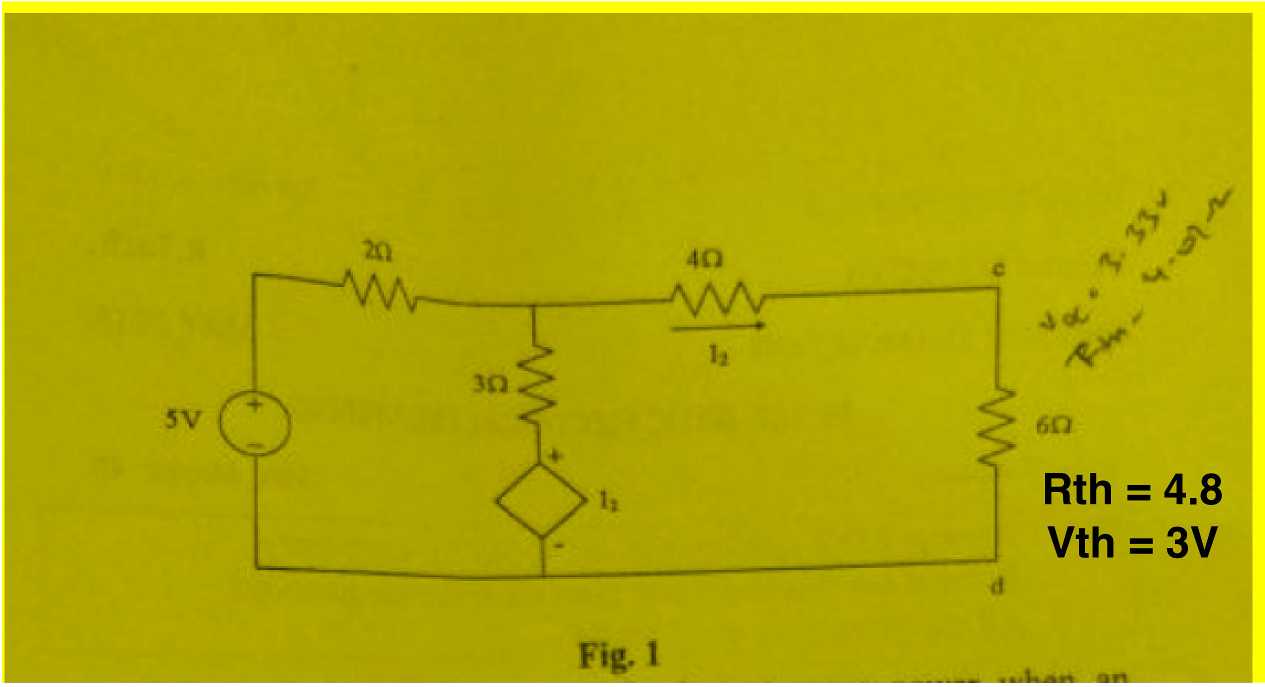
Fig. 4

iv. One of the coils of a transformer is shorted while ... excited. The input current drawn is in phase with respect to applied voltage. (2x4)

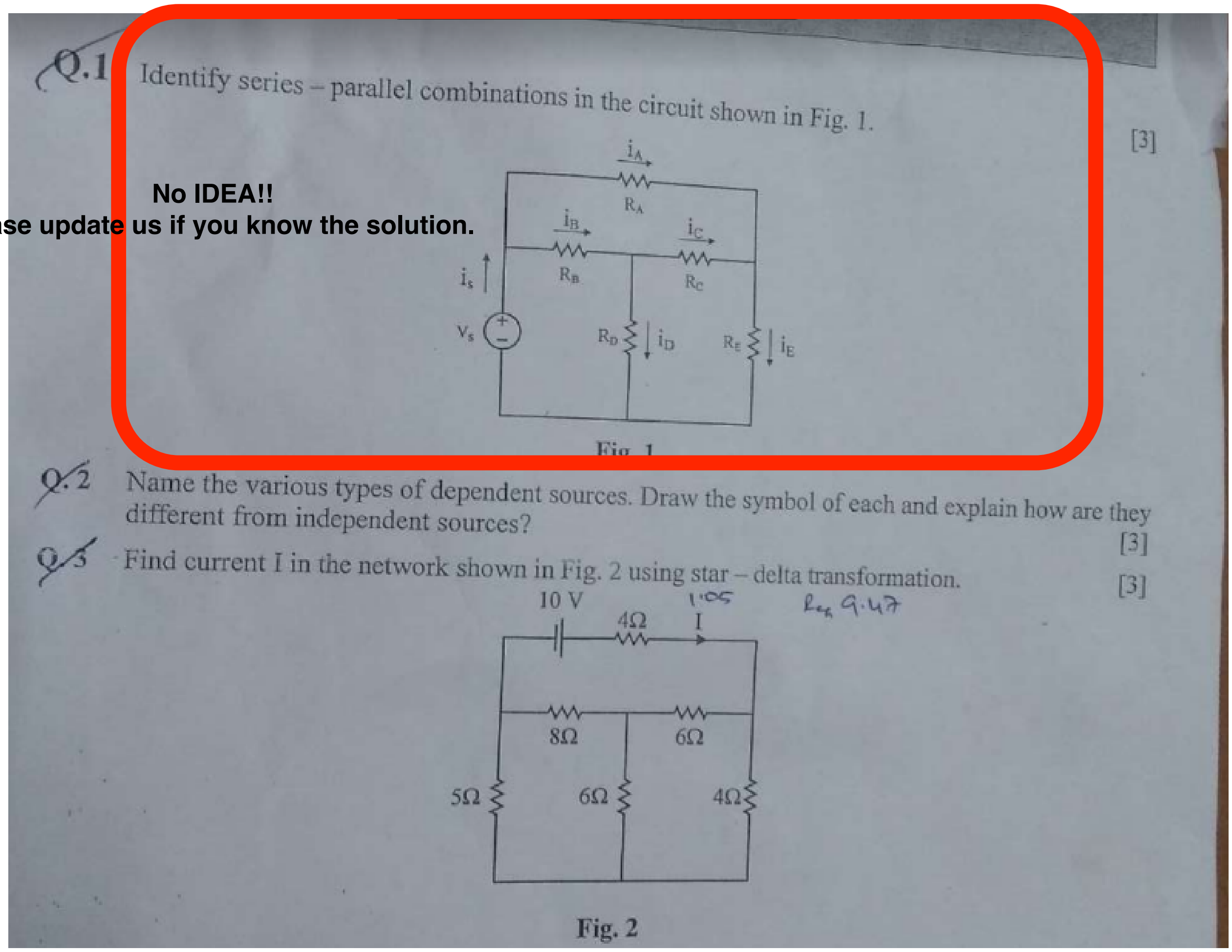
Q. 2(a) State Norton's theorem. Explain the difference between Norton's and Thevenin's theorem. (4)

(b) Determine the Thevenin equivalent circuit as seen from terminals cd for the circuit shown in Fig.1. (4)

P.T.O



Rth = 4.8
Vth = 3V



Q.1 Identify series – parallel combinations in the circuit shown in Fig. 1. [3]

No IDEA!!
please update us if you know the solution.

Fig. 1

Q.2 Name the various types of dependent sources. Draw the symbol of each and explain how are they different from independent sources? [3]

Q.3 Find current I in the network shown in Fig. 2 using star – delta transformation. [3]

Fig. 2

Q.5 State Superposition theorem. Compute the current in 23 Ω resistor of the circuit shown in Fig. 4 by applying superposition principle. [3]

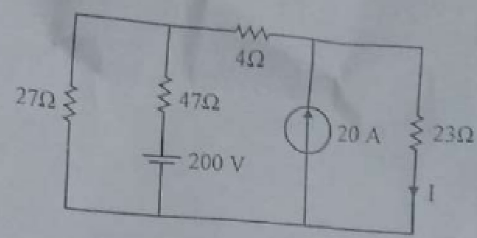


Fig. 4

Q. 6 The voltage and current through a circuit element are

Fig. 5

Q. 9 Find the Thevenin's equivalent of the circuit shown in Fig. 6 as seen from AB.

4 ohm and 10 V

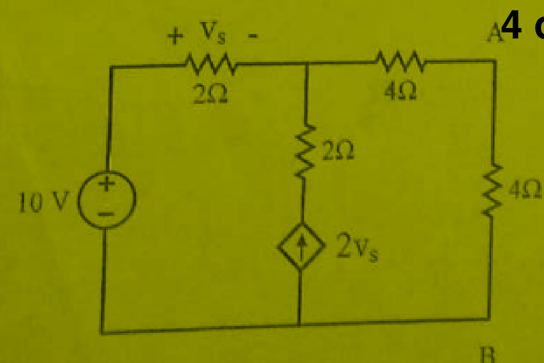


Fig. 6

Q.3. Determine the current through 5Ω resistor in Fig.3 using Nodal Analysis.

− 36.7 LA

3.76 A

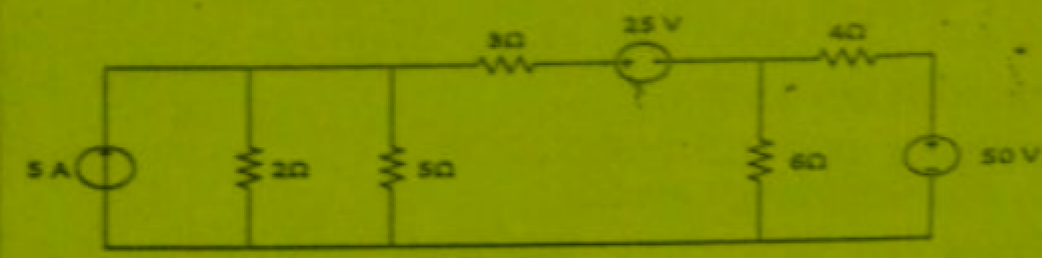


Fig.3

...ing data, if any.

1. Verify superposition theorem for $y = f(x) = 7x + 3$.
2. What will happen if Switch is closed in the following Fig. 1.

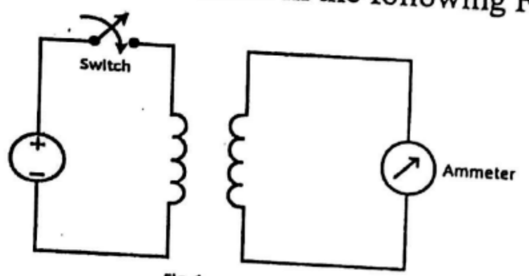


Fig. 1

3. Fig. 2 shows one node of an electric circuit, using KCL, finds V_4. Given $V_2 = 5e^{-2t}$, $V_3 = 2e^{-2t}$ and $i_1 = 2e^{-2t}$

ANS : e^-2t

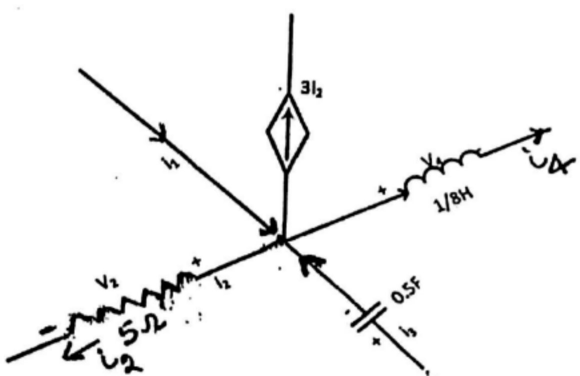


Fig. 2

4. An